1st Tutorial Examination

Course – 315; Time; 45 min

Marks - 10

Q.1. a) Define colligative properties. Write the name of the colligative properties. 1

b) Depression of freezing point of 1 mole of sucrose and 1 mole of NaCl will not be the same. Give reason. 1.5

Q.2. a) What is activated charcoal? 0.5

b) Write Langmuir and Freundlich adsorption isotherm equation and show their validity by drawing plots. 2

Q.3. a) Define partial molal volume. 0.5

b) Calculate the value of Vm for the mixture of A na B having the following parameters: 2

Mole fraction of A =0.1; MW of A= 60; MW of B=80; Density of the mixture = 0.957g/cm3.

Q.4. a) Draw plots showing validity of Beer –Lambert law. Write the equation of straight line for the plot. How could you calculate unknown concentration from the plot? 2.5

1. (g) Why colligative properties of electrolyte and non-electrolytes are not same?

Colligative properties of electrolyte and non electrolyte are not same because electrolytes undergo dissociation or association.

(i) For dissociation of electrolytes the number of particle is increased. so the colligative properties increased or changed.

(ii) For association of electrolytes the number of particles is decreased. So the colligative properties be decreased or changed.

(iii) For non electrolytes there is no association or dissociation can occur. So that the colligative remain unchanged.

That's why the colligative properties of electrolytes and non-electrolytes are not same.

2. (a) Write equation for Freundlich and Langmuin adsorption isotherm. Show the validity of the law graphically. How could you calculate slope and intercept from the plot.

Freundlich adsorption isotherm:

Freundlich proposed an empirical eqn and was known as Freundlich adsorption isotherm. This equation is as follows:

$$\frac{x}{m} = K P^{\frac{1}{n}} \quad \text{---------- (1)}$$

where,

x = mass of gas adsorbed / amount of adsorption.

m = amount of adsorbent.

P = Pressure

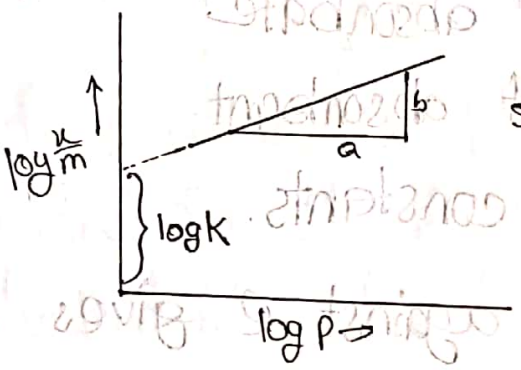
K and n = Constants which depends on the adsorbate and adsorbent.

talking loganitm in eqn (1)

$$\log \frac{x}{m} = \log K + \frac{1}{n} \log P$$

Validity:

If $\log \frac{x}{m}$ is plotted agnist $\log P$, a straight line should be obtained. The slope of the line will give the value of $\frac{1}{n}$ and the intercept on the Y axis give the value of $\log k$



Langmuir adsorption isotherm:

The equation is given as follows,

$$\frac{x}{m} = \frac{k_1 k_2 P}{1 + k_1 P} \quad \text{for gas}$$

again, $\frac{x}{m} = \frac{k_1 k_2 C}{1 + k_1 C}$ for liquid

Now for (i) $\Rightarrow$

$$\frac{x}{m} = \frac{k_1 k_2 P}{1 + k_1 P}$$

or, $$\frac{1}{\frac{x}{m}} = \frac{1 + k_1 P}{k_1 k_2 P}$$

$$\text{or, } \frac{p}{x/m} = \frac{1}{K_1K_2} + \frac{p}{K_2}$$

Similarly for (ii) $\Rightarrow \frac{c}{\frac{x}{m}} = \frac{1}{K_1K_2} + \frac{c}{K_2}$

where,

c = concentration of solute

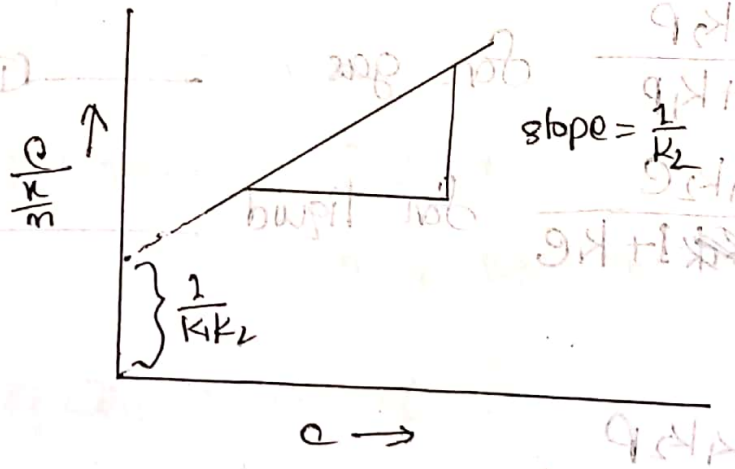
p = partial pressure of gas

x = amount of absorbate

m = amount of absorbent.

K_1 and K_2 are constants.

Validity: A plot of $\frac{c}{x/m}$ against c gives straight line with a slope of $\frac{1}{K_2}$ and intercept of $\frac{1}{K_1K_2}$ and shows the validity of Langmuir adsorption isotherm.



2(b) Define adsonption isotherm:

The amount of adsonbed by any adsonbent is dependent on pressure and temperature.
Thus the amount of adsonbed (x) is a function of pressure (P) and temperature (T) i.e

$$X = f(P,T)$$

If the temperature kept constant and pressure is changed, the curve between X and P is known as adsorption isotherm.

$$X = f(P) \text{ ; T is constant}$$

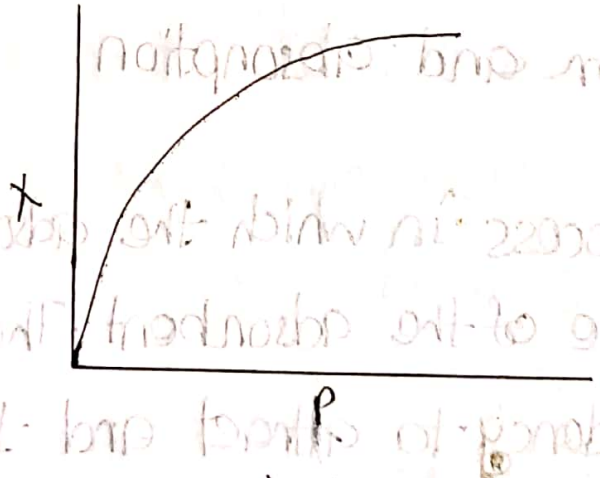


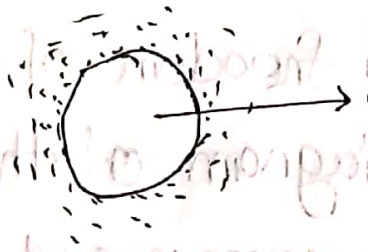
Fig: Adsonption isotherm.

2(c) What is activated charcoal?

Activated carbon also called activated charcoal is a form of carbon processed to have small, low volume pores that increase the surface area available for adsorption or chemical reaction.
Activated charcoal is charcoal that has been treated with oxygen at temperature 665–1125 K to open up the millions of tiny pores between the atoms. Activated charcoal has large number of surface area. So more adsorption occurs.

2(d) Define adsorption and absorption.

Adsorption: The process in which the adsorbate is taking place in the surface of the adsorbent. The surface of a solid has a tendency to attract and to retain molecules of other species with which such surface come in contact. This phenomenon of surfaces is termed as adsorption.



Absorption: Absorption is a bulk phenomenon in which the substance is not only retained on surface but passes through the surface to become uniformly distributed throughout the body of a solid or liquid to form a solution or compound.



absorption process

2(e). Why charcoal in powdered form adsorb more than same amount of solid.

Powdered substance are more effective adsorbent than their crystalline forms because when a substance is powdered then the surface area increases and physisorption is directly proportional to surface area of adsorbent.

3. (a) Calculate the degree of freedom of three component system. Draw the phase diagram of three component system and show the homogeneous and heterogeneous area in the diagram.

According to phase rule the degree of freedom is given below

$$F = e - P + 2 \quad \text{---------- (1)}$$

where, F = degree of freedom

e = No. of component

P = No. of phase

For homogeneous system P=1

from equation (1), $F = 3 - 1 + 2 = 4$

For heterogeneous system P=2

From equation (1), $F = 3 - 2 + 2 = 3$

The phase diagram of three component system is given below-

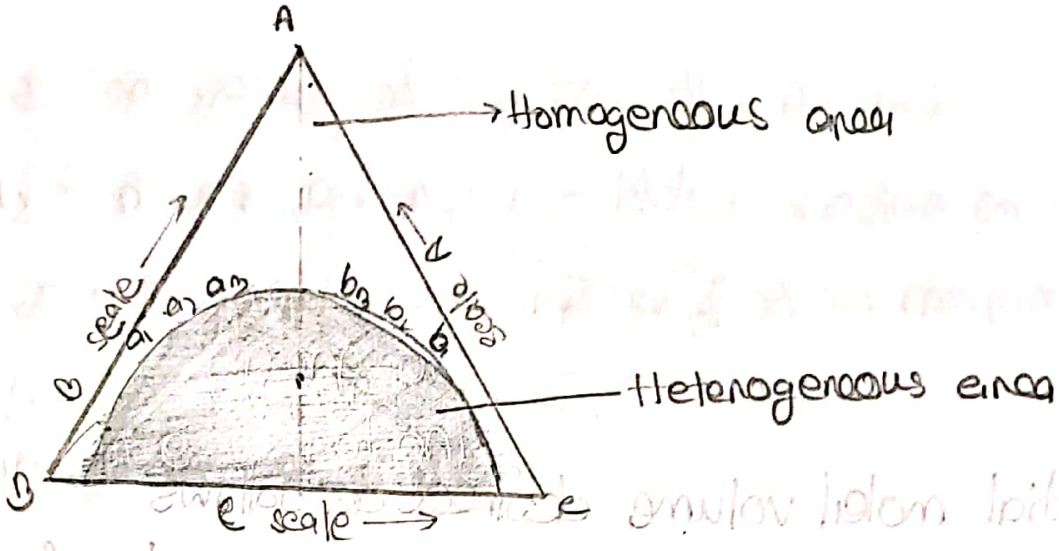


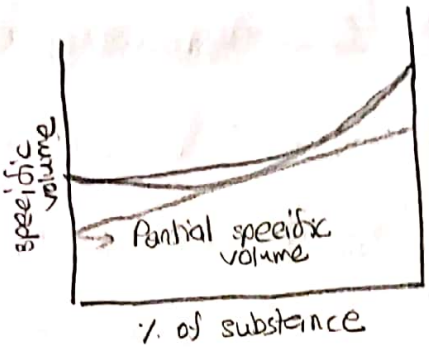
Fig: Phase diagram of 3 component system.

4. (a) Define partial molal volume.

Partial molal volume is the product of partial specific volume and molecular weight of substance. Partial specific vol. is obtained from the plot of specific volume vs % of substance.

Partial molal volume = partial specific volume

× Molecular weight of substance



Partial molal volume defined as follows

Let G represents any extensive property of a binary solution at constant T and P.

The partial molal quantity of these compound is given by,

$$\bar{G}_1 = \left(\frac{\partial G}{\partial n_1}\right)_{n_2, T, P} \quad \text{---(I) Component 1}$$

and $$\bar{G}_2 = \left(\frac{\partial G}{\partial n_2}\right)_{n_1, T, P} \quad \text{---(II) component 2}$$

Total extensive property is given by -

$$G = n_1 \bar{G}_1 + n_2 \bar{G}_2 \quad (T, P \text{ constant})$$

Case of volume,

$$V = n_1 \bar{V}_1 + n_2 \bar{V}_2 \quad (T, P \text{ constant})$$

where, $\bar{V}_1$ and $\bar{V}_2$ are respective partial molal volumes

$$\left.\begin{aligned} \bar{V}_1 &= \left(\frac{\partial V}{\partial n_1}\right)_{n_2, T, P} \\ \bar{V}_2 &= \left(\frac{\partial V}{\partial n_2}\right)_{n_1, T, P} \end{aligned}\right\} \text{Partial molal volume of component 1 and 2.}$$

4(b) Calculate the value of V_m for the mixture of A and B having the following parameters - Mole fraction of A = 0.1 MW of A = 60 ; MW of B = 80 ; Density of the mixture = 0.957 g/cm³

We know,

$$V_m = V_A X_A + V_B X_B$$

$$= \frac{M_A}{\rho} X X_A + \frac{M_B}{\rho} X_B$$

$$= \frac{60 \times 0.1}{0.957} + \frac{80 \times 0.9}{0.957}$$

$$= 81.50 \text{ cm}^3 \quad \text{(Ans)}$$

Here,

$X_A = 0.1$

$X_B = 1 - 0.1 = 0.9$

$\rho = 0.957$ g/cm³

$M_A = 60$; $M_B = 80$

$V_m = ?$

$$V_m = \left(\frac{\partial V_m}{\partial n_i}\right)_{T,P,n_j}$$

$$V_m = V_A X_A + V_B X_B$$

7. (a) Draw plots showing validity of Beer-Lambert law. Write the equation of straight line for the plot. How can you calculate unknown concentration from the plot.

Beer-Lambert's Law states that for a solution, the absorbance (A) of a sample is given by

$$A = \varepsilon c l$$

$$A = \log\left(\frac{I_0}{I}\right) = \varepsilon c l \quad \text{---------- (1)}$$

where, A = absorbance

I_0 = intensity of incident light

I = intensity of transmitted light.

ε = molar absorption

c = conc. of sample

l = path length.

A plot of A against c gives a straight line through the origin.



8. (a) Write equation showing hydrolysis of the ester you have used in the experiment. Write the unit of rate constant (k) for second order reaction.

The hydrolysis of ester (ethyl acetate) by base catalysed is given below:

$$CH_3COOC_2H_5 + OH^- \longrightarrow CH_3COO^- + C_2H_5OH$$

This reaction follows 2nd Order rate law. The unit of rate constant for 2nd order rate law is given below,

we know, rate constant for second order reaction

$$K = \frac{1}{t}\frac{x}{a(a-x)} = \frac{1}{s}\frac{mol\,L^{-1}}{mol\,L^{-1}.mol\,L^{-1}} = L\,mol^{-1}s^{-1}$$

For first order, $K = \frac{1}{t}\ln\frac{a}{a-x} = \frac{1}{s}\frac{mol\,L^{-1}}{mol\,L^{-1}} = s^{-1}$

For zero order; $mol\,L^{-1}s^{-1}$

8(b) Draw necessary plot and show from the plot how you are going to calculate specific reaction rate for the hydrolysis reaction.

The hydrolysis of ester in the presence of base follow 2nd order rate equation. So, the rate constant is given by.

$$K = \frac{1}{t} \cdot \frac{x}{a(a-x)} \quad \text{---(i)}$$

where,

a = initial concentration

x = final concentration.

$(a-x)$ = concentration after time t

when, $t=0$ then c_0 ; $t=t$ then c_t

So, $a \propto (c_0 - c_\infty)$ and $x \propto (c_0 - c_t)$

that means, $a - x = (c_t - c_\infty)$

From (i), $K = \frac{1}{at} \cdot \frac{c_0 - c_t}{c_t - c_\infty}$

$$\Rightarrow c_t - c_\infty = \frac{1}{aK} \cdot \frac{c_0 - c_t}{t}$$

$$\Rightarrow c_t = \frac{1}{aK} \cdot \frac{c_0 - c_t}{t} + c_\infty \quad \text{---(ii)}$$

$$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$$

$$y = m x + c$$

A plot of C_t vs $\frac{C_0 - C_t}{t}$ gives a straight line with slope $\frac{1}{ak}$ and intercept C_∞. Since initial concentration is known, we can calculate the specific rate constant (k) from the slope $\left(\frac{1}{ak}\right)$.

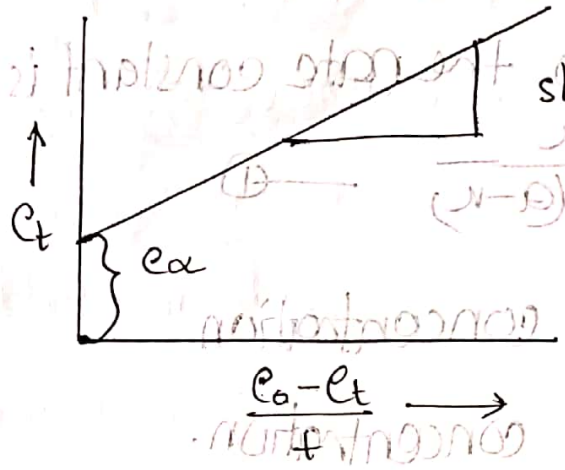


Fig: Plot of C_t vs $\frac{C_0 - C_t}{t}$

8. (c) How could you determine rate constants from conductance measurements?

9. (a) Define molar conductivity. Give unit of molar conductivity.

Molar conductance is defined as the conductance of all the ions produced when 1 mol of an electrolyte dissociates into an ion in a volume V.
It is obtained by multiplying specific conductance and the volume (V) in mL.

$$\Lambda_m = K \times V$$

unit :- $ohm^{-1}\ cm^2\ mol^{-1}$

9. (b) Why equivalent conductance increases with dilution

Equivalent conductance increases with dilution, because it is the product of specific conductivity and the volume V, containing 1 mole of electrolyte.
Although specific conductivity decreases with dilution, this decrease is compensated by the increasing value of the volume.

(*) Equivalent conductance increases with dilution as the degree of ionisation increases with dilution thereby increasing the total number of ions in a solution. Solution which contain large number of ions compared to the other solution of the same concentration at the same temperature has more conductance.

The specific conductivity decreases with dilution as the conductivity of a solution is directly proportional to the no. of ions present in the unit volume of a solution because current is carried forward by the ions, with dilution the number of ions in unit volume decreases. So that conductivity also decreases.

(*) What is degree of dissociation? Write the values of Ke according to ostwalds dilution law.

Degree of dissociation is the fraction a mole of the reactant that underwent dissociation. It is represented by α.

Ostwald's dilution Law is;

"The degree of dissociation of any weak electrolyte is inversely proportional to the square root of the concentration and directly proportional to square root of dilution."

CH_3COOH is a weak electrolyte in water,

$$CH_3COOH_{(aq)} \rightleftharpoons CH_3COO^{\ominus} + H^{\oplus}$$

Mathmatically α is the ratio of Λ_e and Λ_o.

$$\alpha = \frac{\Lambda_e}{\Lambda_o}$$

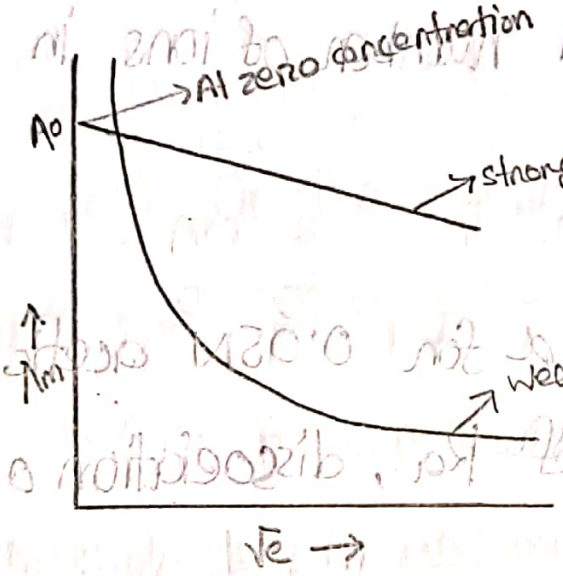
Where, Λ_e = conductance at equivalent

Λ_o = conductance at zero concentration.

The value of Ke according to ostwalds law is.

$$Ke = \frac{\alpha^2 . c}{1-\alpha}$$

9(d) Draw a graph of molar conductivity vs square root of concentration for weak and strong electrolyte and explain the graph:



Molar conductivity of strong electrolyte: There is only a small increase in conductance with dilution. Because a strong electrolyte is completely dissociated in solution and so the number of ions remains constant. At higher concentration the greater inter ionic interactions retard the motion of ions and therefore conductance falls with increasing conc on decreasing dilution.

Molar conductivity of weak electrolyte: Weak electrolytes dissociates much lesser than the strong electrolyte, so

conductance is also lessen/lower. The curve for weak electrolyte shows large increase in conductance ~~measurements~~ with dilution specially near infinite dilution. Because with dilution number of ions in solution increases.

9(c) If the value of α for 0·05N acetic acid is 0·03, calculate the value of K_a, dissociation of acid.

From ostwald's dilution law,

$$K_e = \frac{c\alpha^2}{1-\alpha}$$

$$= \frac{(0{\cdot}03)^2 \times (0{\cdot}05)}{1-0{\cdot}03}$$

$$= 1{\cdot}35 \times 10^{-6}$$

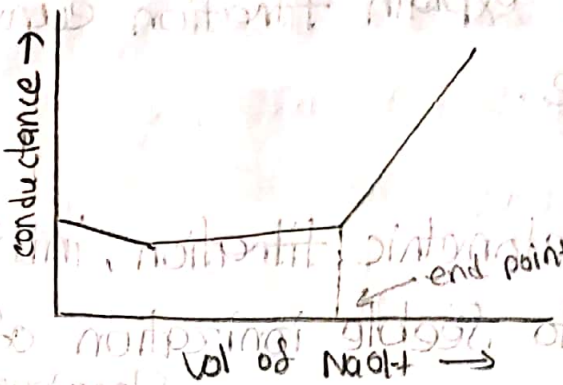
Given that

$c = 0{\cdot}05N$

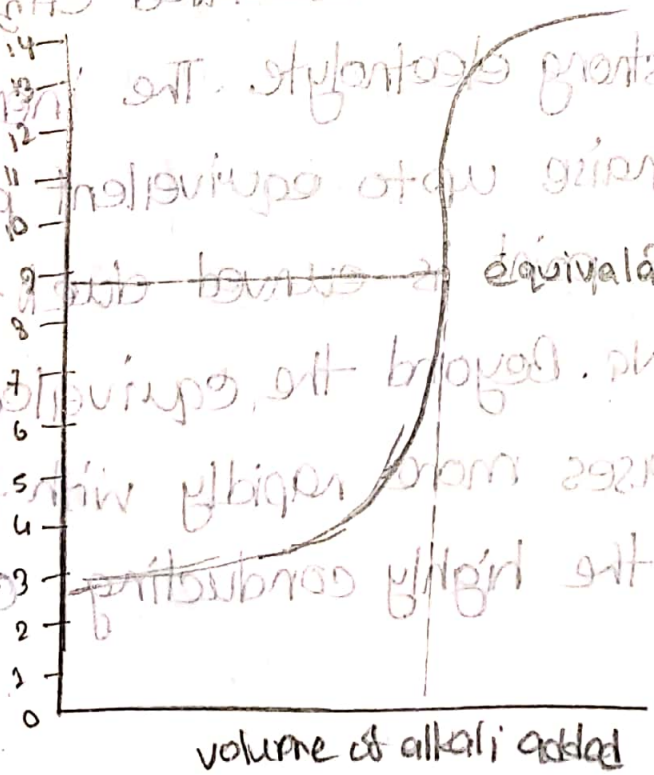
$\alpha = 0{\cdot}03$

(10 + 11) (a) Draw and explain titration curve of weak acid by strong base.

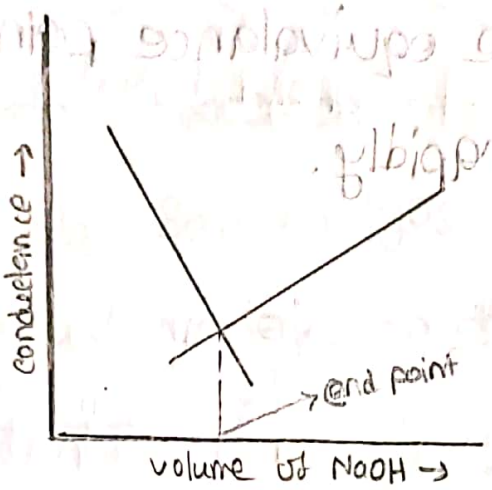
In the case of conductometric titration, initially the conductance is low due to feeble ionization of weak acid (ex. acetic acid). On the addition of strong base like NaOH, there is decrease in conductance not only due to the replacement of H^+ by Na^+ but also surpasses the dissociation of acetic acid (weak acid) due to common ion acetate. But very soon, the conductance increases on adding NaOH as NaOH neutralizes the un-dissociated CH_3COOH to CH_3COONa which is strong electrolyte. The increase in conductance continue to raise upto equivalent point. The graph near the equivalence point is curved due to the hydrolysis of salt CH_3COONa. Beyond the equivalance point, conductance increases more rapidly with the addition of NaOH due to the highly conducting $OH^\ominus$ ions.



In the case of potentiometric titration, it can serve to determine the equivalence point or end point of titration. As the titration is carried out, as long as the acetic acid is in excess, the pH will vary slightly with addition of base. However, near the equivalence point, concentration of the H^+ cation fall rapidly.



(20+21)(b) Draw and explain the titration curve for strong acid by a strong base.



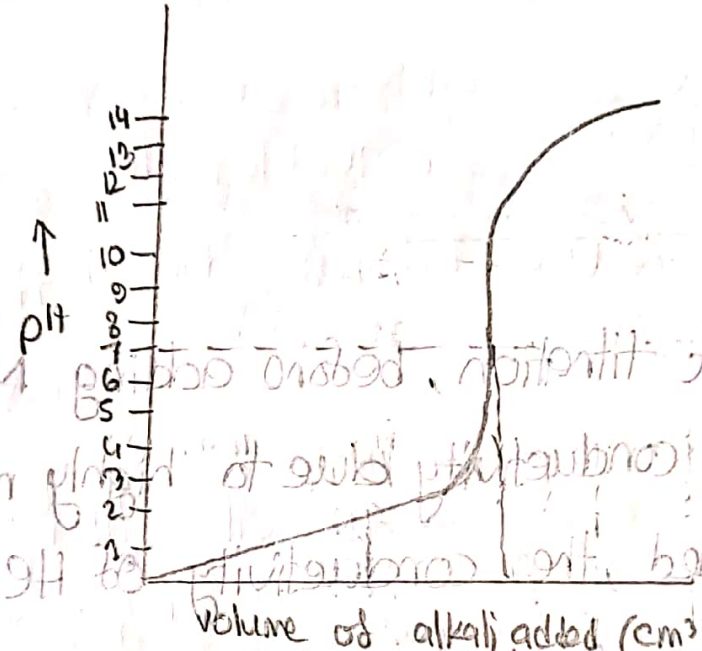
In the conductometric titration, before adding NaOH solution to HCl, HCl has high conductivity due to highly mobile H^+ ions. When NaOH is added, the conductivity of HCl decreases due to neutralization of highly mobile H^+ ions by OH^- ions of NaOH.

$$H^+Cl^- + Na^+OH^- \longrightarrow Na^+ + Cl^- + H_2O$$

The conductivity of solution decreases untill an equivalent point is reached. Further addition of NaOH increases the conductivity of solution due to increase in the highly mobile OH^- ion.

In the case of potentiometric titration, it can serve to determine the equivalence point or end point of titration.

As the titration is carried out as long as the acid is in excess, the pH will vary slightly with addition of base. However, near the equivalance point, concentration of the H^+ ions fall rapidly.



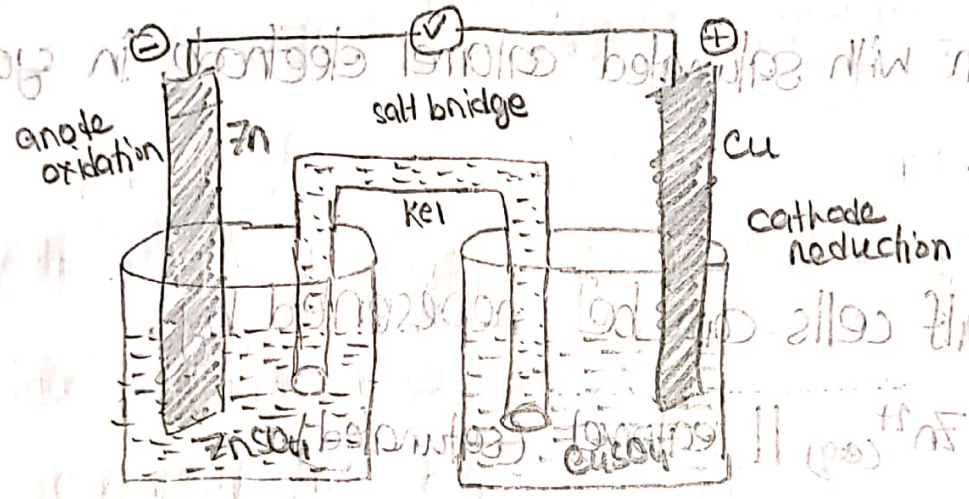
The main function of a salt bridge is given below-

(i) It helps to maintain the electrical neutrality within the internal circuit.

(ii) It also prove

12. (a) What is salt bridge? Why it is used in cells?

A salt bridge is strong electrolyte that facilitates the flow of ions between oxidation and reduction half cells. Example: $AgNO_3$, KCl, etc. Salt bridges are generally used in galvanic cell such as voltaic or daniel cell.



$Zn_{(s)} | ZnSO_{4(aq)} || CuSO_{4(aq)} | Cu_{(s)}$

The main functions of a salt bridge is given below-

(i) It helps to maintain the electrical neutrality within the internal circuit.

(ii) It also prevents the cell from taking its reaction to equillibrium

(iii) It helps preventing the accumulation of positive and negative charges around the respective electrodes and

allowing a smooth reaction to take place.

(iv) It prevents or minimizes the liq-liq junction potential

(v) It acts as a electrical contact between two half cells.

9. (b) While cell notation of the cells prepared by Zn and copper with saturated calomel electrode in your experiment.

(i) Two half cells can be represented by,

$Zn, Zn^{2+}_{(aq)}$ || calomel (saturated)

calomel (saturated) || $Cu^{2+}_{(aq)}, Cu$

The cell notation is,

$Zn_{(s)} | Zn^{2+}_{(aq)} || Cu^{2+}_{(aq)} | Cu_{(s)}$

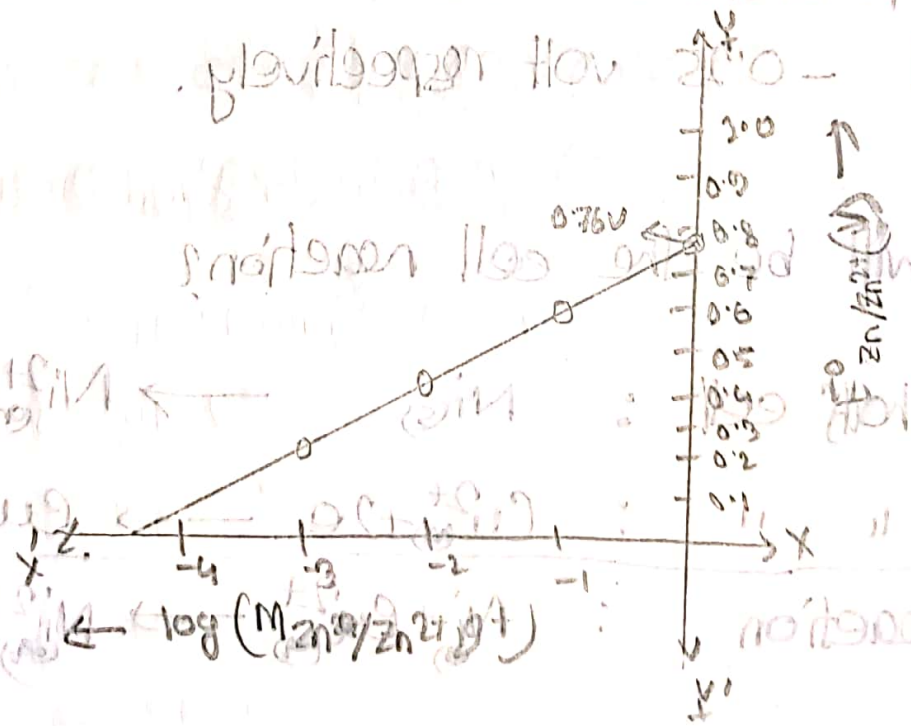
9 (c) Show graphically and from equation how could you calculate standard electrode potential of E° of Zn and Cu.

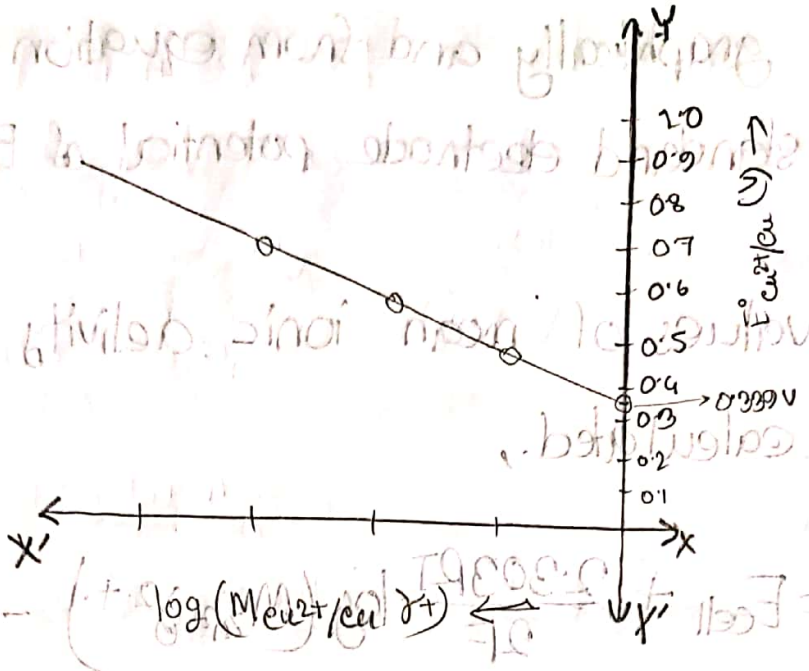
Using the values of mean ionic activity co-efficient ($\gamma_\pm$) E° can be calculated.,

$$E^\circ_{Zn/Zn^{2+}} = E_{cell} + \frac{2.303RT}{2F} \log\left(M_{Zn^{2+}} \gamma_\pm\right) - E_{cal\ (red)}$$

$$E^\circ_{Cu^{2+}/Cu} = E_{cell} - \frac{2.303RT}{2F} \log\left(M_{Cu^{2+}} \gamma_\pm\right) + E_{cal\ (red)}$$

Graphically also E° can be calculated.





9. (a) A cell is prepared by dipping a copper rod in 1M $CuSO_4$ and a nickel rod in 1M $NiSO_4$ solution. The standard reduction potential of Cu and Ni electrodes are 0·34 and −0·25 volt respectively.

⊗ What will be the cell reaction?

Anode half cell : $Ni_{(s)} \longrightarrow Ni^{2+}_{(aq)} + 2e$

Cathod " " : $Cu^{2+}_{(aq)} + 2e \longrightarrow Cu_{(s)}$

Cell reaction : $Ni_{(s)} + Cu^{2+}_{(aq)} \longrightarrow Ni^{2+}_{(aq)} + Cu_{(s)}$

(iv) Calculate the standard EMF of the cell.

$$E_{cell} = E^{\circ}_{cell} - \frac{0.0592V}{n} \times \log Q$$

$$= \{E_{cathode(red)} - E_{anode(red)}\} - \frac{0.0592}{2} V \log \frac{[Ni^{2+}]}{[Cu^{2+}]}$$

$$= 0.34 - (-0.25) - \frac{0.0592V}{2} \log \frac{1}{1}$$

$$= 0.59\,V$$

(v) Which electrode will be positive?

Cathode electrode will be positive, means Cu electrode is positive in this cell.

(vi) How will the cell be represented?

$$Ni_{(s)} | Ni^{2+}_{(aq)} (1M) \,||\, Cu^{2+}_{(aq)} (1M) | Cu_{(s)}$$

9 (e) An electrochemical cell is constructed from silver half cell and copper half cell. The copper half cell contains 0.2M $Cu(NO_3)_2$ and conc. of Ag^+ is 2.6×10^{-7} M. Given that, $E^\circ_{Ag^+/Ag}$ = 0.799V and $E^\circ_{Cu^{2+}/Cu}$ = 0.337 V.

(*) Write cell notation:

$$Cu_{(s)} | Cu^{2+}_{(aq)} (0.1\,M) \,||\, Ag^+ (2.6 \times 10^{-7} M) | Ag_{(s)}$$

(*) Write oxidation and reduction half reaction.

Here copper half cell is anode and silver half cell is cathode.

Half cell reactions are.

Anode half reaction: $Cu_{(s)} \longrightarrow Cu^{2+} + 2e$

Cathode half reaction: $2Ag^+_{(aq)} + 2e \longrightarrow 2Ag$

cell reaction: $Cu_{(s)} + 2Ag^+_{(aq)} \rightarrow Cu^{2+}_{(aq)} + 2Ag_{(s)}$

(*) Calculate the EMF of the cell:

$$E_{cell} = \left(E^\circ_{cathode(red)} - E^\circ_{anode(red)}\right) - \frac{0.0592V}{2} \log \frac{[Cu^{2+}]}{[Ag^+]^2}$$

$$= (0.799 - .337)V - \frac{0.0592}{2} \log \frac{0.10}{2.6 \times 10^{-7}}$$

$$= 0.462 - 0.36 \text{ V}$$

$$= 0.102 \text{ V}.$$

9(i) Half cell reactions for a cell Mg and Sn is as follows.

$Mg \longrightarrow Mg^{2+}(0.01M) + 2e^-$; $E^\circ = 0.34V$

(2M) $Sn^{2+} + 2e \longrightarrow Sn$; $E^\circ = -0.136V$

(ii) Write cell notation,

$$Mg_{(s)} \mid Mg^{2+}_{(aq)}(0.01M) \parallel Sn^{2+}_{(aq)}(2M) \mid Sn_{(s)}$$

(iii) Which electrode will be positive?

Cathode electrode is positive. So Sn electrode will be positive.

(iv) EMF of the cell.

$$E_{cell} = \left\{E_{cathode(red)} - E_{anode(red)}\right\} - \frac{0.0592}{n} \log \frac{0.01}{0.2}$$

$$= \{-0.136 - (-0.34)\}V - \frac{0.0592}{2} \times (-2)$$

$$= 0.024 + 0.0592\ V$$

$$= 0.263V.$$

-1
-2½ Anode.